USN													Course Code:	22MA312/24MA313

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester B.E Degree Examination

Department of Mathematics

ENGINEERING MATHEMATICS-III

(Common to CSE,ISE)

Duration: 3 Hours | Max. Marks: 100

Instructions:

1. Answer all the questions in PART – A.
2. Questions in PART – B have internal choice.

PART – A

Q.NO	1	CO	1	CD	L2	PI	1.1.1	MARKS	10

Fit a curve of the form $y = ax^2 + bx + c$ for the following data, also estimate y when x = 6

x	1	2	3	4	5
y	10	12	13	16	19

Q.NO	2	CO	2	CD	L3	PI	1.3.1	MARKS	10

(a) There are 10 students of which 3 are graduates. If a committee of 5 is to be formed, what is the probability that there are (i) only 2 graduates (ii) at least 2 graduates.

(b) A bag contains three coins one of which is two headed and other two are normal and fair. A coin is chosen at random from the bag and tossed four times in succession. If head turns up each time, what is the probability that this is a two headed coin.

Q.NO	3	CO	3	CD	L2	PI	1.2.1	MARKS	10

(a)The discrete probability distribution of X is given below:

X	0	1	2	3	4	5	6	7
$P(x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

(i) Find the value of k ii) Evaluate $P(x < 6)$, $P(3 < x \leq 6)$

(b)The probability that a news reader commits no mistake in reading the news is $1/e^3$. Find the probability that on a particular news broadcast he commits (i) only 2 mistakes (ii) at most 3 mistakes

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Q.NO	4	CO	4	CD	L1	PI	1.2.1	MARKS	10

(a) How many permutations are there for the eight letters *a*, *c*, *f*, *g*, *i*, *t*, *w*, *x*?

(i) How many starts with the letter *t*? (ii) How many letters start with *t* and end with the letter *c*?

(b) How many positive integers *n* can we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want *n* to exceed 5,000,000?

Q.NO	5	CO	5	CD	L2	PI	1.2.1	MARKS	10

Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes C_1, C_2, C_3, C_4, C_5, one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2, T_3 and T_4 for C_4 or C_5, and T_5 for C_3 or C_4 or C_5. In how many ways can the teachers be assigned the work (without displeasing any teacher)

PART – B

Q.NO	6	CO	1	CD	L3	PI	1.2.1	MARKS	10

Fit an exponential curve of the form $y = ae^{bx}$ for the following data.

x	5	6	7	8	9	10
y	133	55	23	7	2	2

OR

Q.NO	7	CO	1	CD	L3	PI	1.2.1	MARKS	10

Compute the coefficient of correlation and hence obtain the equation of the lines of regression for the following data

x	1	2	3	4	5
y	2	5	3	8	7

Q.NO	8	CO	2	CD	L2	PI	1.3.1	MARKS	10

(a)State and prove Baye's theorem.

(b) Three major parties *A*, *B C* are contending for power in the elections of a state and the chance of their winning the election is in the ratio 1 : 3 : 5. The parties *A*, *B C* respectively have probabilities of banning the online lottery 2/3, 1/3, 3/5. What is the probability that there will be a ban on the online lottery in the state? What is the probability that the ban is from the party *C*?

OR

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Q.NO	9	CO	2	CD	L1	PI	1.2.1	MARKS	10

(a) If A and B are independent events, show that the events
(i) $\bar{A}$ and $\bar{B}$ are also independent. (ii) A and $\bar{B}$ are also independent.
(iii) $\bar{A}$ and B are also independent

(b) If A and B are events with $P(A) = \frac{1}{2}, P(A \cup B), = \frac{3}{4}$ and $P(\bar{B}) = \frac{5}{8}$, then find $P(A \cap B)$, $P(\bar{A} \cap \bar{B}), P(\bar{A} \cup \bar{B})$, $P(\bar{A} \cap B)$

Q.NO	10	CO	3	CD	L2	PI	1.2.1	MARKS	10

(a) From a sealed box containing a dozen apples it was found that 4 apples are perished. Obtain the probability distribution of the number of perished apples when 3 apples are drawn at random.

(b) In 800 families with five children each how many families would be expected to have (i)either 2 or 3 Boys (ii)at most 2 Girls.

OR

Q.NO	11	CO	3	CD	L3	PI	1.2.1	MARKS	10

(a) A continuous random variable has the probability density function
$f(x) = \begin{cases} kx^2, 0 < x < 3 \\ 0, \quad Otherwise \end{cases}$ (i) Find the value of k ii) Evaluate P(1< x < 2)),

(b) If x is normal variate with mean 30 and standard deviation 5 find the probability that
(i) 26 ≤ x ≤ 40) (ii) x ≥ 45, Given $\phi(0.8)$=0.2881, $\phi(2) = 0.4772$, $\phi(3) = 0.4987$
where $\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{\frac{-z^2}{2}} dz$

Q.NO	12	CO	4	CD	L1	PI	1.2.1	MARKS	10

(a) Find the coefficient of $w^2 x^2 y^2 z^2$ in the expansion of $(2w - x + 3y + z - 2)^{12}$
(b) Determine the number of positive Integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5.

OR

Q.NO	13	CO	4	CD	L1	PI	1.2.1	MARKS	10

(a)Determine in how many ways can the letters in **ARRANGEMENT** be arranged so that (i) there are exactly two pairs of consecutive identical letters
(ii) at least two pairs of consecutive identical letters.

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Q.NO	14	CO	5	CD	L1	PI	1.3.1	MARKS	10
Find the number of derangements of 1, 2, 3, 4?									
OR									
Q.NO	15	CO	5	CD	L3	PI	1.3.1	MARKS	10
Using the exponential generating function, find the number of ways in which four of the letters in the word **ENGINE** be arranged									

Course Outcome (CO):	
CO 1	Apply least square method to fit a curve for the given data and evaluate the correlation coefficient and regression lines for the data.
CO 2	Determine the nature of the events and hence calculate the appropriate probabilities of the events.
CO 3	Classify the random variables to determine the appropriate probability distribution.
CO 4	Apply the techniques of counting to identify the number of ways in which a given task can be accomplished without list all the possibilities explicitly
CO 5	Derive the generating function for the given situation and evaluate the required coefficient.

Performance Indicators:	
1.1.1	Apply mathematical techniques such as calculus ,linear algebra and statistics to solve problems.
1.2.1	Apply laws of natural science to an engineering problem
1.3.1	Apply fundamental engineering concepts to solve engineering problem

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyze
L5	Evaluate
L6	Create

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